



The emergence of sign–tool hybridity in situated geometry practice: a networking perspective

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Abstract

A design-based research project evaluating novel resources for culturally situated embodied geometry learning occasioned an opportunity for the researchers to consider conceptual relations between two prominent neo-Vygotskian theories, the semiotic–cultural theory of objectification by Luis Radford and instrumental genesis by Pierre Vérillon and Pierre Rabardel. We examine the case of Nath, an 11-year-old student who spontaneously created *sign–tool hybridity* in the form of a single artifact functioning fluidly as either a semiotic means of objectification or a cognitive instrument supporting his inquiry and argumentation. From an ecological standpoint, the hybridity arose through a perception–action coupling in which the artifact’s meaning and utility co-evolved with the unfolding cultural activity. Analyzing the artifact exclusively as either semiotic or instrumental may compromise the synthetic coherence of an enactivist theoretical modeling. Interfacing epistemologically affiliated theoretical perspectives on the ontology and cognitive function of educational resources may enable researchers to better understand the resources’ rich pedagogical affordances and, thereby, render these affordances accessible to students. Moreover, practicing epistemologically pluralist learning environments, where students are left to tinker with available resources into new semiotic–cum–instrumental objects of their own device, in turn may afford researchers likewise to tinker with available theories toward greater explanatory power.

Keywords Embodied geometry · Instrumentation · Networking theories · Presymbolic notion · Reasoning · Semiotic means of objectification

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1 Overview and objective: articulating the implicit role of pedagogical design in networking educational theories

It could just be that many of us educational researchers are conceptual prisoners of our own design. At least, despite our averred intellectual expansiveness, design-based researchers often construct learning environments whose empirical evaluation yields data inadvertently upholding our own theoretical perspectives. This evidentiary circularity is not disingenuous—we genuinely invest in implementing pedagogical visions in pursuit of instructional excellence. Still, what we get from our research study is *a priori* constrained by what we put in it: students’ observed behaviors may be implicitly formed by our expectations as instantiated in the resources and activities we have created. Is there a way out of this Gordian knot?

One way out, we will argue, is to loosen our grip on the learning environment itself. Rather than scripting tightly structured interactions, we propose creating spaces for students to engage with materials and peers in spontaneous creative forms that are not exhaustively prescribed by the designer. When students are left to tinker materially with mathematical objects—through embodied exploration, manipulation, invention, and social interaction—they are less likely to exhibit behavior that ratifies one theory or another. Instead, such open-ended activity settings invite theoretical surprises: students may unexpectedly exhibit behaviors that align with constructs from multiple frameworks, nudging researchers toward more pluralistic stances that negotiate complementary or even divergent theoretical perspectives.

To illustrate this proposal, we present a design-based research evaluation of novel geometry learning resources that provides an empirical site to examine two epistemic functions of artifacts—semiotic functions as signs, and instrumental functions as tools—by networking two prominent theories in mathematics education: (1) the semiotic-cultural theory of objectification (Radford, 2000) and (2) instrumental genesis (Vérillon & Rabardel, 1995). Conveniently for our alignment efforts, both frameworks are grounded in Vygotskian cultural-historical psychology,¹ which foregrounds the mediating role of artifacts in the development of higher psychological functions. Although these theories share Vygotskian foundations, they diverge in their emphases: the theory of objectification highlights semiotic means of objectification, the culturally and historically sedimented forms through which perception and meaning-making are socially organized; instrumental genesis, by contrast, focuses on utilization schemes and the intertwined processes of instrumentalization and instrumentation through which learners transform artifacts into instruments of activity. Yet, this distinction, we will argue, offers productively overlapping lenses for understanding how students appropriate and repurpose educational resources.

Our study is situated within the broader intellectual strain of *networking theories* in mathematics education, which offers a methodological precedent for connecting diverse perspectives—shaped by distinct epistemologies, methods, and research questions—without collapsing their distinctions (Artigue & Mariotti, 2014; Artigue et al., 2009; Prediger & Bikner-Ahsbals, 2014). Networking seeks productive engagement by maintaining theoretical integrity while enabling cross-fertilization and dialogic synthesis. It represents a middle path between two extremes: on one end, the isolationist stance of disregarding alternative

¹ While *instrumental genesis* initially emerged as a hybrid framework of Vygotsky’s concept of mediation with Piaget’s notion of *scheme*, the present study follows its Vygotskian lineage, emphasizing activity, mediation, and the cultural-historical development of instruments (see Section 3.2).

theories; on the other, the homogenizing pursuit of a unified meta-theory that subsumes individual frameworks at the expense of their distinctive identities (Prediger et al., 2008). Various networking strategies have enabled scholars to clarify relationships among theories (Bosch et al., 2017; Scheiner et al., 2024), apply multiple perspectives to empirical data (Rodríguez-Nieto et al., 2023; Tabach et al., 2020), and develop new frameworks that integrate elements from diverse traditions (Antonides et al., 2025; Artigue et al., 2009).

Guided by this networking orientation, we adopted a *combining and coordinating* strategy (Prediger et al., 2008), fostering mutual understanding between frameworks and exploring their joint use to explain specific learning phenomena. Our coordination operates as a pragmatic bricolage (Prediger & Bikner-Ahsbahr, 2014): we use both semiotic–cultural theory of objectification and instrumental genesis lenses in making sense of empirical data to account for both the cognitive and cultural dynamics of artifact-mediated mathematical activity. As our analysis progressed, empirical patterns suggested that the two frameworks in question—though historically distinct—are sufficiently compatible to warrant a deeper level of theoretical articulation. Our empirical findings reveal a mediational continuity between signs and tools that neither framework, on its own, fully anticipates. We extend this line of thinking by proposing the construct of *sign–tool hybridity*, by which students spontaneously determine and alternate between semiotic and instrumental functions of material resources. Drawing on the functional dynamic systems (FDS) perspective (Shvarts & Abrahamson, 2023), we model this coordination not as a static dual property of artifacts but as a self-organizing and mutually constitutive process, emerging through perception–action coupling within collaborative activity, in which the functions of artifacts shift fluidly with the evolving demands of the task. As students appropriate and reconfigure available media to transform the environment, they reorient their peers’ attention and progressively build their mathematical argumentation.

In what follows, we first introduce the broader context of our project, describe the design of our learning environment, introduce our participants, and outline our methodological approach (Section 2). We then lay out the two theoretical frameworks we seek to network: the semiotic–cultural theory of objectification and the theory of instrumental genesis (Section 3). The “Results and Discussion” section then accounts for several vignettes that illustrate how our data revealed the need to interleave the two theories (Section 4). Finally, in the “Conclusion” section, we offer some closing thoughts on the reciprocity of theory and design (Section 5).

2 Context: design-based research of culturally situated embodied design

2.1 Background

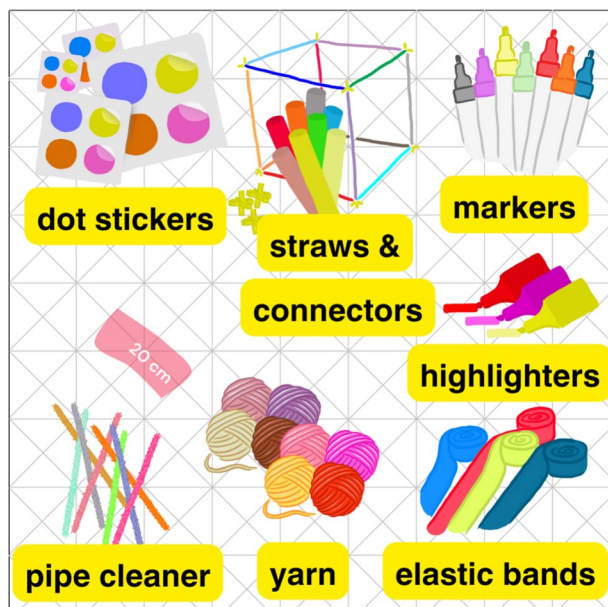
This study is part of a doctoral design-based research project that leverages students’ culturally authentic heritage practices as situated contexts for developing formal mathematical discourse. An experimental activity design evaluated in this study sought to foster students’ perceptuomotor dance competency—specifically, the tacit imaginary projection of line configurations onto their enactive landscape—as a cognitive resource for producing auxiliary constructions in geometric proof. This design is rooted in, yet elaborates on, the embodied-design framework for grounded conceptual understanding (Abrahamson, 2014; Abrahamson et al., 2020) and, more broadly, on research into the movement-based cognitive constitution of academic concepts (Macrine & Fugate, 2022).

We situated our study in local traditional dance in an attempt to leverage students' *funds of knowledge* (Moll et al., 1992)—the repertoire of practices they acquire through active participation in their cultural community. Whereas dance and geometry are traditionally considered disparate cultural practices, Abrahamson and Shulman (2019) have theorized that mathematics and dance share underlying cognitive structures, particularly the construct of the *attentional anchor*—a perceptual cue that guides an agent's movement or action (Hutto & Sánchez-García, 2015). Dancers are known to generate mental imagery to coordinate their bodily movements, especially as choreographic complexity increases (Clark & Ando, 2014). For instance, a Balinese dance student in training created a triangle as her attentional anchor to coordinate her foot configurations across different spatial positions as she followed a choreographed sequence (Apsari & Abrahamson, 2025b). Similarly, in an embodied mathematics classroom, Abrahamson and Bakker (2016) found that students working with the Mathematics Imagery Training for Proportion developed attentional anchors to maintain proportional relationships between their hands in accordance with task constraints. In the present study, our design aims to elicit attentional anchors that mediate student enactment of dance movements as a resource for their mathematical reasoning. We conjecture that, given appropriate conditions, these practices could be mutually constructive. For a detailed account of our cognitive-anthropological domain analysis, which presents the design rationale from a cultural perspective, see Apsari and Abrahamson (2025a).

2.2 Materials

The core objective of operationalizing our design conjecture is to blend students' dance and geometry experiences as laminated in the same spatial and socio-cognitive arena, thus occasioning and legitimizing full-body geometric discourse. For this purpose, the first author created *GRiD*—Geometry Resources in Dance (Apsari & Abrahamson, 2026), a

Fig. 1 Geometry resources in dance (GRiD) resources: floor mat and marking accessories



Note. Graphic artwork: Made Ariawan

diagrammatic floor mat for dance–geometry activities (see Fig. 1) for students to utilize, *if* they so wish and *as* they so wish, during their problem-solving activities.

2.3 Participants

A first pilot test was run in California with a Grade 5 student who had experience in traditional Balinese dance practice. We then implemented the experimental design at an elementary public school in Bali, Indonesia, with 76 Grade 5 students. In total, we worked with 77 volunteering student participants aged 9–12. In general, all Balinese students have some experience in dance practice because it is part of their mandated elementary school program. Notwithstanding, a demographic survey distributed before the study began revealed that 12 of these 76 volunteering participants identified themselves as individuals with no training in Balinese traditional dance. We were not able to ascertain why they marked themselves as such, but suspect they may have interpreted the question as inquiring whether they pursued dance also outside of school settings.

All 76 volunteering participants were grouped into two settings: the 64 students who had stated they had had dancing experience were formed into groups of four or five students each; the 12 remaining students were distributed individually, so that the instructor could guide them to explore the basic postures and movements of the Balinese dance, all vital for the designed learning process. Our case-study student, Nath, was from the dance-experienced group. He worked with three of his classmates, namely Owen, Merry, and Philip (all names are pseudonyms).

2.4 Procedure

We evaluated our design through a *teaching experiment* with the target participants. This method was chosen to provide us, as researchers, with firsthand access to how students' mathematical ideas emerged during the activity, while simultaneously supporting their conceptual development (Steffe & Thompson, 2000). In our study, all students participated in a 60-min activity led by the first author, who is native Balinese and therefore speaks Bahasa Indonesia, Bali's official language. At the beginning of the session, the first author introduced the activity's resources (Fig. 1). Next, students were invited to share their dancing experiences by performing on the mat several basic postures and moves from their local choreographic tradition. This paper focuses on the following specific dance activities within the topic of angle measurement and conservation (see Apsari & Abrahamson, 2026):

1. In Activity 1, students are asked to perform Tapak Sirang Pada (TSP), the Balinese-dance basic foot posture (see Fig. 2a). In this posture, the heels touch while the feet are rotated out, which, in Euclidean geometry, is said to create a 90° angle. Our instructions do not specify the angle that the feet configure in the posture. Only once students are standing in Tapak Sirang Pada do we ask them to evaluate the angle their feet configure.
2. In Activity 2, the students are asked to work in pairs to perform face-to-face Tapak Sirang Pada postures, where their toes meet (see Fig. 2b). The students are then asked to name the quadrilateral they have configured together.

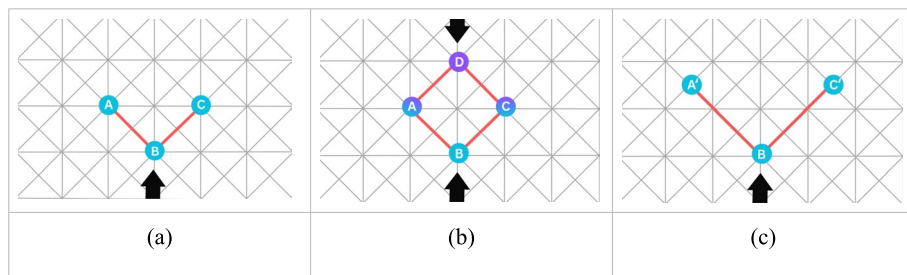


Fig. 2 Dancing geometry problems related to the topic of angle: **a** illustration of the Tapak Sirang Pada basic dance posture on the GRiD mat—left foot (BA) and right foot (BC) configure a 90° angle aperture, **b** two students standing opposite each other in Tapak Sirang Pada posture (ABC and CDA) form a quadrilateral, and **c** Tapak Sirang Pada feet positions of a hypothetical person whose feet are longer than the average child (A'BC')

3. In Activity 3, the students are asked to consider a hypothetical situation, where someone with longer feet than their own would like to learn how to perform TSP (see Fig. 2c). Would these large feet still configure an angle of 90° ?

The key challenge of these tasks lies in angle conservation (cf. Piaget, 1941), which is reported as a common struggle for young students. Activities 1 and 2 are inspired by Bütüner and Filiz (2017) and Devichi and Munier (2013) who found that students struggle to recognize a mathematical object when it is presented in non-canonical form—compare the V-shaped right angle in Fig. 2a with the L-shaped right angle often found in textbooks; the students who participated in those studies apparently believe that a form-maintaining change in a geometrical figure's orientation may change its angle measurements. Further, Activity 3 targets another naive belief that has been repeatedly documented in the literature, namely that an angle's measure changes with the length of its rays (Ozen Unal & Urun, 2021; Sari et al., 2021).

As the study participants performed the Balinese dance postures, the first author guided them to reflect on how they were evaluating and monitoring the quality and accuracy of their performance; students were encouraged to discuss these issues with their peers. In the course of this conversation, participants were invited to use any of the available resources, including a variety of material multimedia and embodied multimodal semiotic resources, such as actual enactment, pantomime, gestures, speech and other vocalizations, text, and drawings. The expectation was that students would spontaneously recognize in the available material resources embedded utilities for enhancing the enactment, explanation, or evaluation of their actions and that, in so doing, they would implicitly appropriate mathematical frames of reference, per the intervention's objective, thus adopting a mathematical discourse of embodied movement (Abrahamson, 2014; Abrahamson et al., 2011).

2.5 Data collection

The teaching experiment took place in an available spacious room in the school facility. Participants and the researcher sat on the floor with their shoes off, as is normative in the Balinese community's traditional dancing culture. Before the activity began, the researcher invited the students to explore GRiD's various artifacts and accessories. The researcher also reminded

students of their study-participant rights to ask for a break or to stop at any time during the session. With the students' and their parents'/legal guardians' approval, the activities were recorded with two audio–video cameras that captured the scene from different angles. We also collected photo evidence of students' artifacts, including their markings, drawings, and writings on the GRiD as well as any constructions they created with the accessories.

2.6 Data analysis

Five native Bahasa Indonesia speakers, who are also fluent in American English, translated and proofread all transcription data into English. The comprehensive data were then analyzed according to principles of micro-genetic ethnography by examining for moment-by-moment learning that may have occurred during the activity (Siegler, 2006). We therefore attended to participants' multimodal utterances, gestures, actions, and material productions and transformations. In addition, following the ethnomethodological approach to conversation analysis (EMCA), we examined the interactional function of students' utterances and actions as their means of maintaining mutual intelligibility (Mondada, 2019). Finally, following guidelines of collaborative micro-ethnography (Schoenfeld et al., 1993), the authors attempted to enhance the reliability of their interpretations and validity of their claims by further discussing selected data vignettes with each other as well as with a group of graduate students who all work on embodied design.

Over a period of 6 months, the research team engaged in extended collaborative micro-genetic analysis of the participants' behaviors. This protracted discourse gradually homed in on several moments of interest in the video data corpus that stood out to us as paradigmatically manifesting students' different socio-cognitive intentionalities, actions, and utterances pertaining to the available material resources. As we examined these moments, our analytic discussions repeatedly drew on constructs from the theory of objectification (e.g., sign-mediated collective meaning making) and instrumental genesis (e.g., tool-oriented schemes of use), as each appeared useful for interpreting aspects of the same episodes. What was not immediately clear, however, was how these theoretical lenses might cohere analytically, particularly in episodes where a single child's activity could be interpreted, across moments, through either construct.

Recognizing our recurring semiotic–instrumental interpretive fluctuation, as we analyzed students' behaviors, compelled us to seek coherence across the two theories in question. Rather than framing our inquiry as “Theory #1 *or* Theory #2,” we asked what it would mean to pursue “Theory #1 *and* Theory #2” and what meta-conceptual quality such an *and* might bear. This question led us to a networking process that followed a strategy of *combining and coordinating* theoretical lenses (Prediger et al., 2008). While prior studies have demonstrated the value of mobilizing multiple perspectives to enrich analysis (e.g., Gellert, 2008; Maracci, 2008), our aim extended beyond alternation toward consolidation. To this end, we sought to identify shared epistemic grounds across the two frameworks whereby the child's two behaviors we had been modeling as disparate were in fact two states of one and the same basic drive. As elaborated in Section 4, our analytic expedition focused us on the constructs of a *sign* (semiotic–cultural objectification) and *tool* (instrumental genesis), and these, in turn, led us to consider the notion of a *sign–tool* as a proposed Vygotskian dualism.

3 Two theories as lenses to unfold mathematical reasoning

Two theoretical frameworks from the learning sciences literature emerged in the course of our microgenetic analysis as potentially informing a deeper understanding of students' task-oriented activity centered on the material media, namely their spontaneous strategies and modes of engagement in utilizing these media as various means of coordinating joint action and promoting collaborative argumentation: the semiotic–cultural theory of objectification by Luis Radford, and the theory of instrumental genesis by Pierre Vérillon and Pierre Rabardel (1995). Toward explaining how these theories are networked in and through our data, this section presents each of the theories.

3.1 Semiotic–cultural theory of objectification

Luis Radford's work frames mathematical meaning making as a culturally and historically situated, socially embodied process mediated through diverse semiotic modalities and forms, such as gestures, artifacts, and language. In developing his *semiotic–cultural theory of objectification*, he engages critically with two major philosophical traditions: C. S. Peirce's pragmatism and Edmund Husserl's intentionalist phenomenology (Radford, 2006). Peirce conceptualizes thinking as a process rooted in a triadic relationship among signs, objects, and interpretants, where meaning emerges through the habitual effect of signs on human interpretation. In contrast, Husserl emphasizes subjective meaning making and intersubjectivity but gives limited attention to the cultural and historical conditions through which shared meanings emerge. While both traditions offer valuable insights, Radford maintains that they fall short in explaining how individuals come to understand mathematical concepts through participation in culturally situated practices. To address this limitation, Radford draws on phenomenological philosophy as well as on Vygotsky's cultural-historical theory of cognitive development to propose a semiotic–cultural perspective. According to this semiotic–cultural theory of objectification, meaning is simultaneously subjective—as intended by the individual—and culturally mediated, emerging through historically grounded semiotic activity (Radford, 2021).

Within this framework, *objectification* refers to the process by which a mathematical “object,” writ large, becomes perceptible to the learner (Radford, 2003). Crucially, objectification does not simply involve recognizing an object, but encountering it in a qualitatively new way. Radford (2002) illustrates this through van Gogh's recollection of his art teacher, who guided him to perceive familiar scenes with fresh insight. In mathematical reasoning, where objects are often abstract and intangible, objectification is essential—the construct of objectification articulates how students first encounter or reconfigure mathematical structures, often for the socio-discursive purpose of engaging in collective reflection upon the objects.

Whereas objectification is a cognitive act, it does not transpire in some would-be Platonic ontological void. Instead, objectification is embodied and instantiated in enactive, at times material concretization. Radford (2003) introduces the concept of *semiotic means of objectification*—the signs, tools, and communicative resources that mediate students' engagement with mathematical content. These semiotic means span material artifacts (e.g., rulers, calculators), linguistic resources (e.g., words, metaphors), as well as spatially constituted signs, whether inscribed or dynamic (e.g., gestures, figures, inscriptions, or, for that matter, computer-based models). Through the coordinated use of these resources, students

signify their covert attention into the public social arena, thus enhancing opportunities to contribute to participating in shared mathematical activity.

In keeping with Radford's framework, we do not treat learning as the cerebral internalization of propositional knowledge but as apprenticed participation in culturally situated semiotic activity. From this systemic perspective, mathematical thinking is not an internal cognitive process but intrinsically a socially organized discursive form of perceiving and acting coupled with signs. Students' (collective) engagement is mediated by culturally available resources, and their mathematical activity unfolds iteratively as they draw on these resources to participate in historically grounded ways of making sense. For example, in algebraic patterning tasks, students' reasoning often burgeons and elaborates through depictive gesture and spatial adumbration as much as through symbolic notation (Radford et al., 2007).

In our own study, we adopt the theoretical construct of objectification as an analytic lens to examine how students construct and communicate mathematical ideas—such as angle, angle measurement, and angle conservation—through collaboratively negotiating the coordinated use of language, embodied signs, and physical tools in the service of accomplishing their task, such as determining the geometrical properties of diagrammatic forms or collectively configuring a target shape. Whereas the theory of objectification foregrounds the semiotic–cultural processes through which mathematical objects become individually and socially perceptible, our second theoretical lens, instrumental genesis, highlights functional dimensions of students' interactions with (material) artifacts rendered as applied utensils promoting task performance. Positing instrumental genesis as theoretically complementary to the semiotic–cultural perspective frames our comprehensive dual analysis of how signs (as semiotic means of objectification) may come to serve as tools (and back again, implementing new operational invariants) within the loose flow of students' mathematical activity.

3.2 Instrumental genesis

Pierre V rillon and Pierre Rabardel's theory of instrumental genesis centers on the mediating role of artifacts in human cognition and action. According to this perspective, subjects (i.e., human agents) come to understand the world through their interactions with the environment, including material artifacts (and the case has been extended to digital and other artifacts). In their triadic neo-Vygotskian/neo-Piagetian model, these Artifacts mediate the relationship between the Subject and the Object of their action. Once an artifact—a human-made material object—is appropriated and integrated into a subject's activity, it becomes more than a tool; it becomes a psychological construct: an instrument (V rillon & Rabardel, 1995).

The process through which an artifact is transformed into an instrument is called *instrumental genesis*. This process involves the development of a complex, evolving relationship among the Subject, the Object of action, and the Artifact itself (V rillon & Rabardel, 1995). A classic example is the case of an infant learning to use a spoon. Initially, the spoon is merely an artifact, but through the baby's efforts to grasp and manipulate it—for instance, to scoop milk or mashed potatoes—it gradually becomes an instrument shaped by the task and the user's developing schemes. Historically, the concept of instrumental genesis emerged as a hybrid Vygotskian–Piagetian construct, rooted in activity theory but borrowing Piaget's notion of scheme to model the user's cognitive assimilation of the instrument (V rillon & Rabardel, 1995). Subsequent educational elaborations situated

instrumental genesis within semiotic and social mediation frameworks (Bartolini Bussi & Mariotti, 2008; Béguin, 2007; Rabardel & Samurçay, 2001; Shvarts et al., 2021), consolidating its cultural-historical orientation.

Rabardel (1995) identified two interrelated processes within instrumental genesis: *instrumentalization*, referring to the applicative transformations, temporary or permanent, that subjects bring to artifacts, and *instrumentation*, referring to the changes in the subject's cognitive schemes as a result of interacting with the artifact. These processes are dynamic and reciprocal. At times, this interaction may result in *catachresis*, where an artifact is used in ways that exceed or diverge from its intended design (Rabardel, 1995). Such repurposing is not considered a misuse, but rather a sign of the subject's creative engagement and interpretive agency. Designers may propose particular uses for an artifact, but users ultimately determine how it is appropriated and integrated into their activity (Rabardel, 2001).

A growing body of research in mathematics education has applied the theory of instrumental genesis to investigate how mathematical knowledge is constructed through students' interaction with instruments. Some examples could be seen in the use of computer algebra systems to explore graphs and functional properties (Artigue, 2002), calculators to investigate limits (Trouche, 2005), and interactive digital embodied tools to learn trigonometry (Shvarts et al., 2021). These studies underscore that integrating technology into the classroom is not simply about providing students with tools, but about opening up new forms of mathematical engagement. Student understanding emerges through both the constraints and affordances of the artifact and through the ways students adapt and appropriate these tools for their emerging purposes.

Importantly, instrumental genesis does not occur in isolation. Rather, it is shaped by students' interactions with peers, teachers, and the broader learning environment (Trouche, 2020). Teachers can support and guide students' instrumental genesis through *instrumental orchestration*—the deliberate structuring of learning activities and classroom interactions to foster productive use of artifacts (Drijvers et al., 2010; Trouche, 2004).

In the context of our study, instrumental genesis offers a theoretical lens for understanding how our student participants appropriate and transform everyday materials into contextually meaningful mathematical implements, that is, concrete resources spontaneously applied as manipulable utensils vital for promoting collective geometric argumentation. For example, when students work with straws and connectors to identify a right angle and then transport that material angle form across space to support an argument about angle conservation under transformation, we observe how the artifact's function evolves through students' interaction with the materials and each other vis-à-vis the task at hand. This instrumental perspective complements our semiotic lens to theorize students' perceptual, symbolic, and operational actions, a theme we develop further in our analysis, below.

4 Results and discussion

Prior to gathering empirical data to evaluate our experimental design's activities, we had never planned to even consider these conceptual relations between the theories. Rather, we were motivated to do so only in the course of analyzing the work of one group of four students—Merry, Nath, Owen, and Philip—who will be the focus of this paper. The group was selected due to their problem-solving strategies which, we came to appreciate, involved first objectification and then, still using the same material resource, instrumentation. We

selected an episode of problem-solving activities of the focus group that we presented into vignettes to exemplify how we combine and coordinate the two theories.

Section 4.1, “Students’ objectification of an angle measurement,” reports on our analysis of how the group identified the angle formed by two feet in a Tapak Sirang Pada posture and the angles inscribed in a quadrilateral built from paired Tapak Sirang Pada postures. The episode culminates in students using plastic straws to accentuate the angle at their joined heels. We interpret these manual placements as students’ appropriation of available semiotic means to objectify a presymbolic notion of 90° .

In Section 4.2, “Instrumentation and instrumentalization as a means of argumentation,” the students are still debating over the quadrilateral’s angles. Therein, one student, Nath, adapts his pair of straws, connected by a joiner node, to lift and transpose the composite structure onto the angle under discussion as a shared procedure for determining whether it too measures 90° . We regard Nath’s action as exemplifying instrumental genesis—how an artifact comprising straws and connectors became an instrument to measure angle equivalence through the intertwined process of instrumentation and instrumentalization.

Section 4.3, “Elaborating on the theoretical construct of sign–tool hybridity,” synthesizes the students’ evolving use of straws in Sections 4.1 and 4.2 to demonstrate how semiotic and instrumental functions can co-develop in a collaborative mathematical problem-solving space. Finally, Section 4.4, “Designing for learning, designing for researching,” steps back to the multi-aspectual design elements that enabled such behaviors and argues for their pedagogical value.

4.1 Students’ objectification of an angle measurement: from a 2-feet dance posture to a 90° angle form

The original design had anticipated that participants would have completed evaluating their individual Tapak Sirang Pada posture angle aperture (see Fig. 2a) prior to constructing a shape composed of two postures (see Fig. 2b). In practice, however, several student groups examined the double-Tapak Sirang Pada posture shape *prior* to determining the single posture angle. The consequent conceptual complexity of evaluating a quadrilateral consisting of unknown angles required creative actions, as we now exemplify with our focus group.

Merry (M), Nath (N), Owen (O), and Philip (P) debated whether the constructed shape (see Fig. 2b) was a square or rhomboid. To begin with, they agreed to assume that their feet were of equal length, so that two students could, at least in principle, configure a rhombus. They then discussed the similarities and differences between a square and a rhombus, eventually homing their inquiry onto the angles in the shapes they had created. If all four of a quadrilateral’s angles are 90° , they argued, the 2TSP figure would be a square (and therefore *not* a rhombus²). A cognitive–developmental factor complicates the students’ exploration and debate—they do not yet “conserve” the right angle when it is rotated out of its canonical vertical/horizontal orientation (what we will call the L-shape 90° ; see Piaget, 1941).

The group began their inquiry by evaluating the angle formed by each TSP ($\angle ABC$ and $\angle ADC$ in Fig. 2b). They carefully assess the position of their feet, which are marked with dot stickers connected by lines. Owen proposed an embodied geometric argumentation

² While a square is a special type of rhombus, these students’ current grade level has not yet encountered that form of geometric reasoning. Hence, for the purposes of this study, we will respect our participating students’ classification convention: a rhombus is a quadrilateral that is equally sided but not equally angled, because the latter property would make it a square.

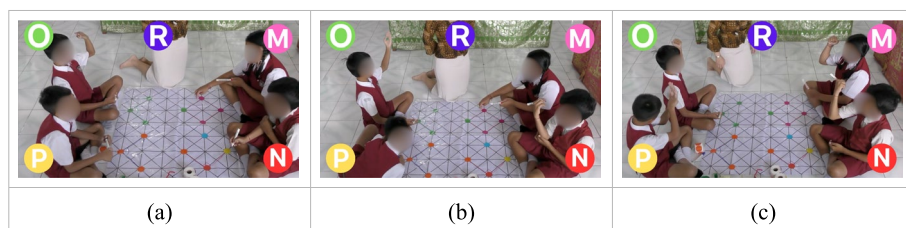


Fig. 3 Embodied geometric argumentation: “Use your arm!”—Owen, looking to his left arm, suggests a strategy to distinguish **a** acute, **b** obtuse, and **c** right angles, with the three other group members following and mimicking

(Palatnik et al., 2025) by using his arm as a tool to measure the angle. According to him, the classification of an angle as either acute, obtuse, or right can be determined by observing how wide he needs to open his folded-back forearm so as to match the angle in question.

As he was demonstrating his arm strategy (Fig. 3), Owen uttered “like this” (informal Bahasa Indonesia, *gini*) and “this much” (*segini*). The qualitative quantifier “this much” suggests that Owen perceives this set of three static postures not as a nominal sequence but as extents (degrees) of rotation. That is, when Owen moves his forearm from one pose to the next, he does so not just as an analog means of iteratively displacing it to its successive locations—this act of moving is itself significant to enacting and gauging the angular aperture. Owen’s coordination of gesture, action, and speech may thus be interpreted as a *semiotic node*, in which multiple semiotic means work together to achieve knowledge objectification (Radford, 2009).

Owen then attempted to instrumentalize his arm as a means of measuring the Tapak Sirang Pada posture angle aperture as marked on the mat (Fig. 4a–b). Contorting his shoulder, elbow, and hand, he struggled to appropriately laminate his arm-as-protractor onto the mat markings, eventually conceding that this embodied tool was unwieldy and ultimately inadequate for the particular context of application. Scanning the mat for alternative means of gauging the angle measure, Owen eyed Philip’s marker. He borrowed the marker and placed it next to his own marker, so that they share a vertex of sorts, thus creating a two-pronged loose construction (the markers lay adjacently, touching at the vertex, without any firm element affixing them solidly into a single rigid unit, Fig. 4c).

Observing Owen’s work, the first author reminded the group that they are welcome to use any available resource. Immediately, Owen picked up two plastic straws and placed them inside his TSP in lieu of the two markers, respectively aligned with his feet. Note that

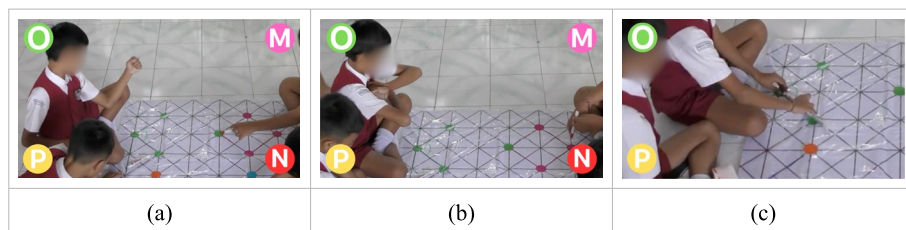


Fig. 4 Owen’s attempt to classify the angle of the Tapak Sirang Pada posture he had marked on the mat: **a** using one arm, tucking his elbow in, **b** forming the angle out of two forearms, and **c** using two markers

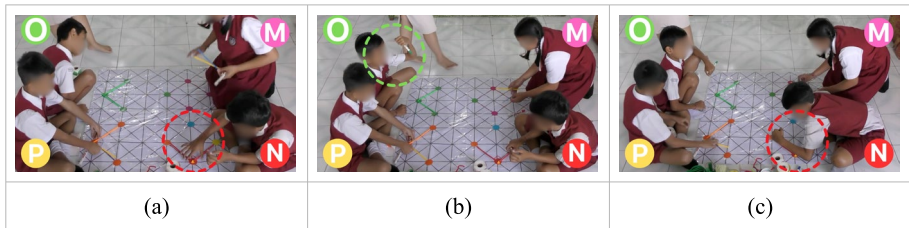


Fig. 5 Straws are introduced into the problem space as new semiotic means, alternative to the marked grid lines and/or instead of the markers that had been placed thereupon: **a** Nath investigating $\angle BAD$, an angle within the quadrilateral construction composed of two opposing Tapak Sirang Pada postures; **b** Owen is suggesting to use the arm strategy for assessing the V-shaped angle; and **c** Nath executed Owen's idea by placing his forearm on the mat. We highlight the main action in **(a)**, **(b)**, and **(c)** with a dashed circle

these straws, too, served essentially to highlight the TSP form—they, too, were not affixed together. Owen's peers quickly imitated him, placing two straws upon their own V-shaped TSP markings.

After placing his straws on his Tapak Sirang Pada feet, Nath moved them to $\angle BAD$ (Fig. 5a, recall Fig. 2b for reference). He was curious about what angle that would be, even though the group had not concluded the Tapak Sirang Pada angle ($\angle ABC$), and asked for Owen's opinion (Fig. 5a). Now “armed” with his two straws on the mat (Fig. 5b) as proxies for his arm and forearm, Owen reintroduced to the group his earlier proposal to use the arm–forearm strategy as a means of determining the TSP angle. Prompted by Owen's suggestion, Nath placed his arm–forearm 90° L-shaped embodied form onto the mat to evaluate whether or not $\angle BAD$ is a right angle (Fig. 5c). Intently comparing his arm–forearm angle gauge with the straws he joined on the mat, Nath concluded that the angle must be 90° , and ended up disagreeing with Owen and Philip's claim that the angle is acute. The conversation about $\angle BAD$ then shifted back to the preliminary purpose of investigating the TSP angles $\angle ABC$ and $\angle CDA$. Only after concluding that $\angle ABC$ and $\angle CDA$ are both 90° angles (see Apsari & Abrahamson, 2026) did the group turn their attention to the other two angles ($\angle BAD$ and $\angle BCD$).

4.2 Instrumentation and instrumentalization as a means of argumentation: from a 90° angle form to a portable measurement device

While all four group members utilized a pair of straws to mark their TSP form and objectify the 90° configuration, Nath elaborated on this disjoint two-straw structure by using a plastic connector to conjoin the straws at the Tapak Sirang Pada angle vertex (Fig. 6a, cf. Merry's *unconnected* straws to Nath's right, Fig. 6c). As we shall see, this modest feat of engineering will soon open new actionable horizons.

Supplemented with the connector, Nath's two-straw structure now bears a new utilization scheme for his geometrical inquiry into both determining the posture's angle measurement and, simultaneously, identifying the mystery quadrilateral constituted from two Tapak Sirang Pada postures. This new structure bears a key feature—it carries 90° as an inherent integral property. The veracity of this 90° property draws both on the geometric form of the connector itself, whose tines are orthogonal, and from the figural match with the gridlines. At no point did any of Nath's peers contest the structure's would-be right angle, and so we assume that this property was implicitly endorsed by his peers. Newly

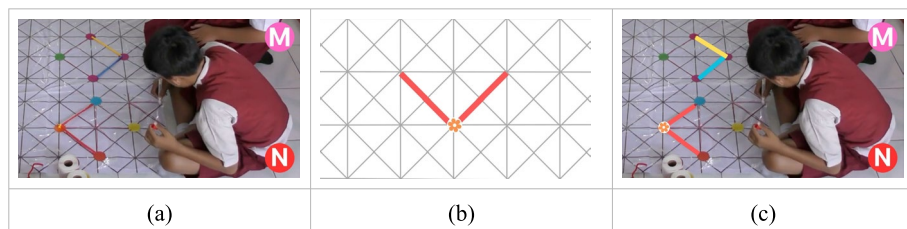


Fig. 6 Nath's straw construction was created by joining two straws with a connector: **a** Nath is building a two-straw + connector construction of a 90° angle; **b** we henceforth use a flower and two rays **c** to depict these constructions (cf. a loose structure of two unconnected straws, to Nath's right)

equipped with this structure of his own device, Nath's perception of the shared geometric phenomena being examined—the *domain of scrutiny* (Goodwin, 1994)—is extended with a publicly adjudicated powerful means of probing line intersections to gauge their angle. Through this process of instrumentation, Nath's contraption of attached straws enlarged the territory of artefactual thinking, enabling a deeper form of objectification. If previously the students had to rotate their heads to align their naked eye with an intersection's L-orientation, Nath's eye now became *domesticated* (Radford, 2010) into new modes of attending to right angles, mediated by an object that itself could be rotated to fit the L.

In Fig. 7, we will observe how Nath uses his straw tool to compare angles at different mat locations. We will also observe that Nath's peers contributed to his inquiry, thus implicitly endorsing his instrumented geometrical strategy. Specifically, Fig. 7e–f features Philip and Owen's verbal input and Mary's hands-on intervention, when Nath was confused about the location of his original TSP.

Our analyses of Nath's diverse applications of his construction have led us to the following interim account. *The material straws' haptic-portable affordances keyed for Nath an emergent ontological dialectics from semiosis (semiotic means of objectification) to instrumentation, whereby the structure's form constituted a hybrid and liminal sign-tool.* This tool, in turn, enabled Nath to gauge for invariance via transitivity: the straw form became imbued with the 90° property of $\angle ADC$ by virtue of tracing it. Next, this property could be exported from $\angle ADC$ to assess $\angle BAD$ and $\angle BDC$. If it fits, then all angles are equal, Q.E.D. Eventually, the group concluded that all four sides of the 2TSP Quadrilateral ABCD are equal as are all its angles, so that the shape is a square.

Whereas the square-or-rhombus dilemma was now resolved, the straw construction's journey continued. Nath's geometrical instrument served prominently in his further exploration of how the straws and their connectors can be a helpful tool for his problem-solving activities. Figure 8 captures the evolution of Nath's straw instrument that iteratively and adaptively served his emerging needs, with each form suggesting a new function, and each function in turn driving a new form.

Following Activity 2, Nath repurposed his two-straw construction into a T-shaped, three-straw construction that he spun idly while awaiting the next task (Fig. 8b). In Activity 3, students considered a hypothetical case of a person with longer feet standing in Tapak Sirang Pada posture, first locating Point C' given Points A' and B, and then assessing whether $\angle A'BC'$ would remain 90° despite its longer rays (see Fig. 2c). To determine Point C', Philip and Nath traced an imaginary horizontal auxiliary line from Point A' until it intersected the other ray (Apsari & Abrahamson, 2026). Point A' was distant from Nath's

 N: Here, here [<i>grabs his straw construction</i>]	 N: [<i>rotates the straw construction to the adjacent side of the quadrilateral</i>] To here. R: Oh, I see. You can check from there. Is that the same as the one at your feet, Nath?	 N: [<i>lifts his straw construction and “translates” it to the opposite side</i>] R: Hmmm... What about on your feet? N: Same R: Where are your feet, Nath?
 N: This [<i>points to his leg</i>]. Eh, this one [<i>points at (2), and then (1), (2), (3)</i>] R: Yes, so where should you put it [<i>refers to the straw construction</i>]?]	 N: Like this? [<i>moves the straw construction to the other side</i>] P: No, no, no! N: Here here [<i>adjust the straw construction to align better with the lines on the GRiD’s mat</i>]	 M: No! Like this, at your feet! [<i>moves the straws to the marked location where Nath’s feet had been originally</i>] P & O: Yes!

Fig. 7 Transcription of straw construction as an argumentation tool

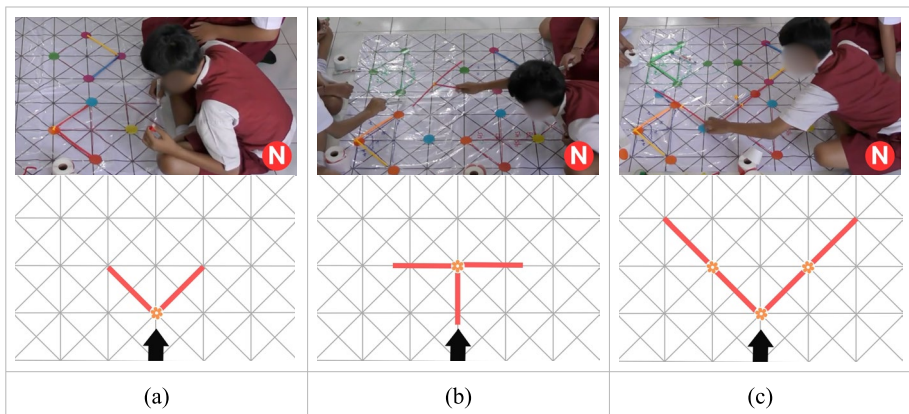


Fig. 8 Adaptive evolution of Nath’s straw construction during a geometry inquiry. **a** Initial V-shaped configuration formed by conjoining two straws with a connector. **b** Transformation into a three-straw T-shaped form, later repurposed to point toward distant locations and trace imaginary lines during collaborative inquiry. **c** Reconfiguration into an extended V-shaped right angle using four straws and additional connectors to accommodate longer rays. Top-row photographs show each stage as enacted; bottom schematics depict the corresponding constructions from a common orientation on the GRiD, highlighting successive structural modifications

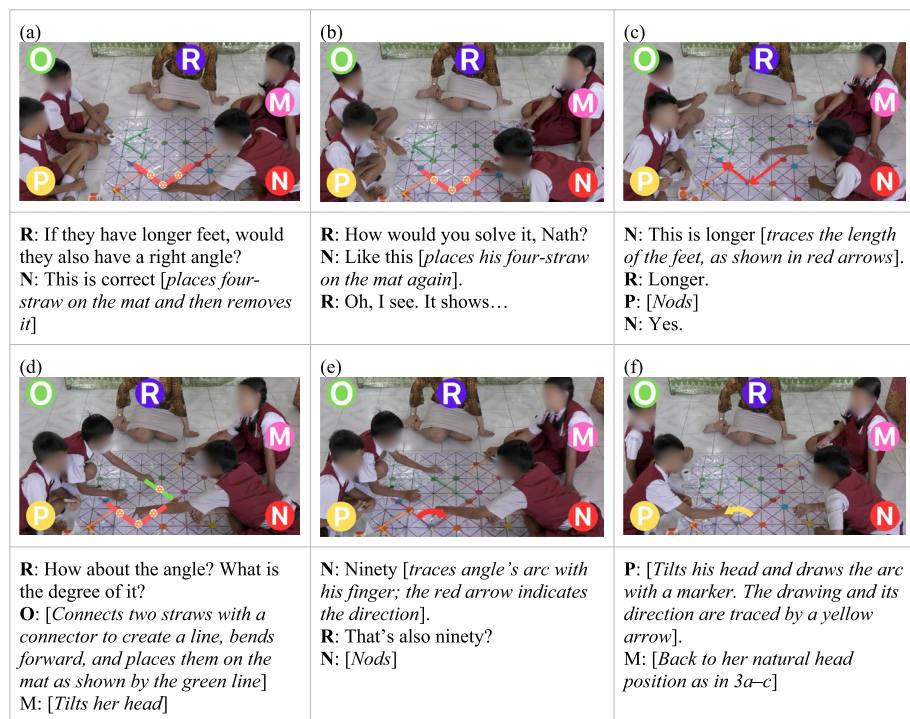


Fig. 9 Transcription of angle measurement does not change when the angle's rays are longer

sitting position, and so he used his three-straw construction as a pointing/tracing tool (Fig. 8b).

Having successfully demonstrated that the rotation of an angle does not change its measure (see Fig. 7), Nath continued to argue that the length of an angle's ray, too, does not impact its measure ($\angle ABC = \angle A'BC'$; see Fig. 8c). To do so, Nath first removed the third straw in the T-shape construction, rendering the construction once more as a two-strawed V-shape, yet then modified it again by connecting one additional straw along each ray, creating a four-straw "Big V" construction. He then placed this four-straw construction on the hypothetical longer-feet Tapak Sirang Pada posture that had been marked on the GRiD to argue that this longer-feet posture has the same measurement as the original 90° one (i.e., the angle enclosed by the original two-straw, which can still be discerned as embedded within the four-straw construction, had never been modified, and so it retained whatever properties it had possessed earlier, ergo 90°).

As Fig. 9 makes evident, Nath employed his Big V four-straw construction to argue that the angle remains unchanged despite its longer rays. Further, Nath's argument that the longer-feet Tapak Sirang Pada posture is a 90° angle was endorsed by the cohort as indicated by their various responses. For example, Merry (Fig. 9d–e) and Philip (Fig. 9f) tilted their heads to perceive a V-shaped angle as a canonically L-shaped—a strategy that the group had often used to evaluate whether an angle was 90° (see Apsari & Abrahamson, 2026). Neither Merry nor Philip remonstrated following the head tilt, suggesting they had sanctioned Nath's assertion.

4.3 Elaborating on the theoretical construct of sign–tool hybridity

Attempting to negotiate their perspectival conflict, our focus group of Merry, Nath, Owen, and Philip employed various semiotic means to argue and communicate their reasoning. The evolution of their embodied semiotic progression—from first enacting the Tapak Sirang Pada foot posture to marking its configuration with stickers and lines, gauging the angle with an arm, appropriating a pair of straws, and then, upon recognizing the 90° structure, returning to the Tapak Sirang Pada posture now newly seen through this right-angle lens—exemplifies a *chain of signification*, in which each successive signifier reorganizes the meaning of the preceding one, which “slides under” the new signifier rather than simply being represented by it (Cobb et al., 1997). Crucially, this transformation is not merely representational. Through the lens of the theory of objectification, these embodied forms and material artifacts function as cultural-historical mediators: they do not mirror pre-existing inner meanings but enable new forms of action and perception. The students’ successive reconfigurations thus constitute an emergent cultural activity in which historically sedimented knowledge—about right angles, bodily alignments, and geometric structures—is refracted and reorganized through their embodied engagement. In this sense, the chain of signification we observe is simultaneously semiotic and praxeological: each new material configuration affords fresh ways of acting mathematically while retaining traces of the cultural practices from which it originated.

While other students in the group employed the straws to mark their feet as they solved the posture problem, Nath was the first in his group to conjoin the straws with a connector. This small connector, a humdrum bit of plastic with minute orthogonal tines, soon changed the course of the group’s mathematical micro-history. By rendering the right angle as a single, fixed composite unit, the connector enabled students to port the angle “as is,” with no apparent concern that it would lose reliability during transport to other locations on the mat.

We argue that Nath’s straw construction constitutes a peculiar ontological hybrid. On the one hand, the artifact draws attention to, and enhances the perceptibility of, a material feature of the environment: the pair of lines marked on the mat to denote the internal angle of the Tapak Sirang Pada posture configuration, itself aligned with the GRiD mat’s line structures during collaborative problem solving. On the other hand, once the group established that the Tapak Sirang Pada posture foot angle is 90° , and the straw construction reliably matched that configuration, the device alternately functioned as a makeshift tool for evaluating angles elsewhere on the mat—even in locations where no feet had ever trod. The evidence of hybrid functions, *sign–tool*, prompted us to revisit the theories of semiotic means of objectification and instrumental genesis, which have historically developed independently, each inheriting Vygotsky’s scholarship that crystallized an ontological distinction between two apparent anthropological functions of artifacts, either as signs or as tools.

Our empirical findings reveal a mediational continuity between signs and tools that neither framework, on its own, fully anticipates, making visible their shared Vygotskian foundation in which signs and tools operate as continuous forms of mediation rather than distinct categories. In cultural-historical forms of situated learning, signs and tools are not mutually exclusive. For example, the theory of semiotic mediation views tools and signs as dialectically intertwined in the development of mathematical knowledge—the use of tools generates signs, and signs reorganize how tools are understood and used

(Bartolini Bussi & Mariotti, 2008). Drawing on both the anthropological theory of the didactic (ATD) and the action–production–communication (APC) framework, Arzarello et al. (2008) theorize that ostensive objects, such as gestures, inscriptions, and material configurations, function not only as signs but also as essential tools of mathematical practice. Building on this foundation, Arzarello and Sabena (2014) further elaborated the systemic and dynamic interplay of these functions through their semiotic-bundle framework, in which tools operate as semiotic resources and signs emerge through their use, reciprocally reorganizing tool-based action.

Further, we extend this line of thinking by proposing the construct of sign–tool hybridity, in which students spontaneously determine and alternate between semiotic and instrumental functions of material resources. The straws’ semiotic and instrumental functions do not appear as fixed categories but as fluidly shifting subjective epistemic roles that may diverge across participants. Such interpersonal variability, which reflects the artifact’s intrapersonal epistemic polysemy, emerges through coordinated inquiry, whereby meaning may stabilize through social negotiation. The co-evolution of these functions in students’ embodied activity can be illuminated through the functional dynamic systems (FDS) perspective (Shvarts & Abrahamson, 2023). Grounded in ecological ontology, FDS views learning as a self-organizing process where cognition emerges from agents’ dynamic coupling of perception and action. In this sense, sign–tool hybridity reflects a functional coordination between semiotic means of objectification and instrumental schemes, a continuous process unfolding as students engage bodily with artifacts. Students appropriate and reconfigure available media to transform the environment, materially and psychologically, thereby reorienting their peers’ attention and sustaining continuity in their mathematical argumentation. This framing suggests an evolutionary continuity from tacit perceptual skills to cultural mathematical practices, with mathematical concepts emerging as situated discursive responses through embodied engagement with artifacts rather than as abstract entities detached from action.

Taken together, our analyses of the vignette’s temporally concatenated episodes show how a semiotic means of objectification became seamlessly instrumentalized as an argumentative utensil, thereby articulating sign–tool hybridity as the conceptual outcome of the networking process enacted in this study. As an outcome of a design-based approach that deliberately left room for material tinkering and collective inquiry, sign–tool hybridity offers a modest but hopefully generative lens for tracing how meaning making and tool use become reciprocally constituted in empirical settings.

4.4 Designing for learning, designing for researching: reflecting on conditions for epistemic instrumentalization

In this section, we step back to ponder various material, cultural, social, and axiological qualities of the interaction that may have contributed to the epistemic instrumentalization of semiotic forms. As such, even as we aim to demonstrate the conceptual interleaving of two prominent theoretical models, we are also noting multi-aspectual design elements that enabled these behaviors, which we evaluate as pedagogically desirable.

Although neither the semiotic–cultural theory of objectification nor instrumental genesis explicitly informed our activity’s design, engineering, or early analysis phases, the learning environment created rich opportunities to examine the cognitive functions of material media. Extending work that initiates theory networking already at the design level (e.g., Artigue & Mariotti, 2014; Polotskaia, 2020) we position the design of the learning

intervention as a site where theoretical perspectives can meet empirically. The activities and facilitation were deliberately designed to support student tinkering, allowing participants to appropriate and transform materials on the fly, reshaping these resources according to their idiosyncratic emergent needs, even beyond the designers' original intent—what Rabardel (1995) characterizes as *catachresis*. Our study shows that theoretically generative moments can emerge not only through post hoc analytical alignment but also from the affordances of the design itself—*when students are invited to tinker with materials, researchers are likewise afforded opportunities to tinker with theory*.

Throughout the teaching experiment, we strove to foster in each student group socio-mathematical norms (Yackel & Cobb, 1996) of multimodal collaborative argumentation (Schwarz & Prusak, 2016), whereby students are encouraged to participate in evidence-based mathematical reasoning that draws on multi-aspectual semiotic resources in dialogue with their peers' consensual, complementary, or contrasting perspectives. Consider Activity 2, where students debated over the identity of the geometrical shape formed by two opposing TSPs. Across groups, some students believed this shape was a non-square rhombus, while others opined that it was a square.

Peers' perspectival conflict over the perception of shared displays may generate conceptually productive dialogue (Benally et al., 2022), because it stokes intellectual need (Harel, 2013), requiring students to construct more rigorous arguments to codify the display based on its evidentiary objective properties (e.g., a square and a rhombus both have four equal sides, but a square also has four right angles). In such accounts, tools and signs function as intertwined means for transforming the environment and making thinking visible (Shvarts & Abrahamson, 2023).

We argue that providing space for such casual mathematical tinkering supports *epistemological pluralism* (Turtle & Papert, 1991), a pedagogical vision that values students' practical know-how, engineering proclivity, and material ingenuity to diversify and democratize participation in STEM practice. Endorsed by the group, Nath's straw construction became a publicly shareable object through which his argument could be examined and refined. Notably, the straws performed an *instrumental* function by enabling the enactment of a 90-degree angle, an *epistemic* function by allowing the group to evaluate angle conservation under transformation, and a *discursive* function by helping Nath articulate, explain, and justify his reasoning to others (see Abrahamson et al., 2011). In fact, as evidenced in Section 4.2, Nath's peers offered him constructive verbal feedback and physical assistance, further stabilizing the construction as a shared epistemic resource.

5 Conclusion

Scholars of mathematics education do not share fundamental assumptions about the nature of the phenomena they investigate: is cognition sequestered to the head, or is it inherently entangled with the material, social, and cultural environment? This absence of a consensual common theoretical ground can so challenge productive dialogue between our constructs and models that we remain in our separate intellectual fiefdoms. Whereas we may each take comfort in the internal coherence of our theoretical frameworks, a persistent reluctance to bridge epistemological divides ultimately undermines our collective humanistic endeavor to improve education through deeper understanding. Over the past decades, mathematics education researchers have explored ways to bring different theoretical frameworks into dialogue through a range of networking strategies.

A recent systematic review by Scheiner and Bosch (2025) highlights that such efforts have often leaned toward *centripetal forces*—that is, toward theoretical unification and integration. However, the authors caution that this tendency should be balanced by *centrifugal forces*, which preserve the distinct contributions of individual theories. Attending to both, they argue, enables more productive and generative theoretical connections.

In response to the transformative call for networking theories, we came to perceive our data as evidencing a hybrid construct, the sign-tool, as instantiating theoretical dialoguing between two important theories in the field of mathematics education—the semiotic-cultural theory of objectification from Luis Radford, and instrumental genesis (or instrumented activity situations) originally from Pierre Rabardel. This dialogue was occasioned in the context of qualitatively analyzing empirical data gathered during the experimental implementation of a novel, culturally situated embodied design for geometry, in which students’ dance-oriented covert attentional anchors were rendered as overt markings on a gridded mat and subsequently enhanced through the placement of colorful plastic straws. While these straws initially functioned to visually amplify the markings, they also introduced new affordances for manual engagement: movable lines that preserved the geometrical properties of the underlying signs. Through this process, the markings became hybrid sign-tool configurations, ultimately consolidated when students attached the straws into a single rigid artifact that could be transported across the mat as a bonafide mathematical instrument.

We find it striking and informative that students can demonstrate such compelling mathematical argumentation using not ready-made artifacts they must master, but their own self-generated thinking-and-communicating instruments. Learning environments that support tinkering and preserve structural looseness are especially productive for surfacing students’ idiosyncratic orientations, enabling diverse perceptions to emerge and be negotiated in a collaborative problem-solving space.

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Author contribution RAA: conceptualization, methodology, formal analysis, investigation, resources, data curation, writing (original draft), writing (review and editing), and project administration; DA: formal analysis, and writing (review and editing).

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Data Availability The data and materials supporting this study’s findings are not publicly available. The data are, however, available from the authors upon reasonable request.

Declarations

Ethics approval Ethical approval for this study was obtained from the Committee for Protection of Human Subjects (CHPS) at the University of California, Berkeley (Federalwide Assurance: 00006252).

Consent for participation and publication Participation consent and media release records were obtained from the participants and their parents/legal guardians prior to the start of the study. The participants and their parents/legal guardians were informed about the purposes of the study and how the participants could voluntarily participate.

Competing interests The authors declare no competing interests.

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